

3D Coordinates Extension — English Worksheet (Beginner)

Topic: Design Your Own 3D Build (x, y, z) · **Course:** 3D Coordinates · **Level:** Extension (Beginner, after Lesson 3) · **Time:** about 35 minutes

Pixel art was flat — two numbers, one wall. This lesson your build is **solid**. Three numbers. You draw it from the **front** and from **above**, then read both drawings back into the world.

Build your canvas. Run this to lay a blank **15 × 15** floor, then put a **red block at your feet** as your **home spot**:

```
def on_run():
    blocks.fill(WHITE_CONCRETE,
                pos(1, 0, 1),
                pos(15, 0, 15),
                FillOperation.REPLACE)
    player.on_chat("run", on_run)
```

Your build stands on that floor. **x** goes across, **y** goes up, **z** goes away from you — a **15 × 15 × 15** space.

● **Stand on the red block every time you run.** If you move, the build moves too.

1 · Predict

Read the steps. Circle your answer.

```
place blue block at (3, 9, 4)
place blue block at (4, 9, 4)
place blue block at (5, 9, 4)
```

A line across, up, or away? Circle one: across · up · away

Which number changes? Circle one: x · y · z

```
place blue block at (7, 2, 4)
place blue block at (7, 3, 4)
place blue block at (7, 4, 4)
```

Across, up, or away? Circle one: across · up · away

Which number changes? Circle one: $x \cdot y \cdot z$

place blue block at (7, 5, 1)

place blue block at (7, 5, 2)

place blue block at (7, 5, 3)

Across, up, or away? Circle one: across · up · away

Which number changes? Circle one: $x \cdot y \cdot z$

2 · Spot the Bug 🐛

This code is broken. Read what it should do, then fill in the fix.

Bug — The table's two legs should sit at **two different spots**: (2, 1, 2) and (2, 1, 4). Both legs land in the **same spot**.

```
place brown block at (2, 1, 2)
place brown block at (2, 1, 2)
```

Hint: the third number (z) of the second leg is wrong. One leg is further **away**, so its z must be **bigger**.

Fill the blank to fix the second line:

```
place brown block at (2, 1, 2)
place brown block at (2, 1, __)
```

Which number fills the blank? Circle one: 2 · 4

Why was the code wrong? Circle one: both legs had the same z · the colour was wrong

3 · Show Your Work 📷 🎥

Stand on your **home spot** (the red block).

Your five rules.

#	Your design must...
1	fit inside the 15 × 15 × 15 space
2	use 3 or more different block colours
3	use 40 or more blocks
4	be recognisable — someone else can name it without being told
5	be drawn on both grids below before you place a single block

Two drawings, one build. The front view gives you **x** and **y**. The top view gives you **x** and **z**. Together they place every block.

Here is one I made — a tree:

🌳 **Front view — x across, y up**

	1	2	3	4	5
5		🌿	🌿	🌿	
4		🌿	🌿	🌿	
3		🌿	🌿	🌿	
2			🌳		
1			🌳		

🌳 **Top view — x across, z away from you**

	1	2	3	4	5
5					
4		🌿	🌿	🌿	
3		🌿	🌳	🌿	
2		🌿	🌿	🌿	
1					

Read it back. The trunk is at $x = 3, z = 3$, and stands from $y = 1$ to $y = 2$. The leaves fill $x = 2$ to 4 and $z = 2$ to 4 , at $y = 3$ to $y = 5$.

Now draw yours. Colour the squares.

Then build it. Read the front view for **x** and **y**. Read the top view for **x** and **z**. Place each block at its (x, y, z).

Record **one video** — one take, no stopping (a phone is fine). Show these in order:

1 • Your grids. Hold this paper up to the camera.

2 • Your build. Point the camera. Hold your paper next to the build so both are in the shot.

Say these out loud, filling in the blanks:

Today I built _____.

I built it using this code: _____.

In this code I used _____.

The hardest part was _____.

That part was hard because _____.

The most fun part was _____.

Something new I learned was _____.

Submit 

Send this worksheet + your video to teacher on KakaoTalk.